

HOIP Deficiency (Immunodeficiency-115 with Autoinflammation): A Comprehensive Disease Characteristics Report

Target disease: HOIP Deficiency · **MONDO:** MONDO:0957981 · **Gene:** *RNF31* (HOIP) · **Category:** Mendelian, autosomal recessive inborn error of immunity

Summary

HOIP deficiency is an ultra-rare, autosomal-recessive inborn error of immunity caused by biallelic loss-of-function or severely hypomorphic mutations in **RNF31**, the gene encoding **HOIP** (HOIL-1-interacting protein), the catalytic RING-IBR-RING (RBR) E3 ubiquitin-ligase subunit of the **Linear Ubiquitin chain Assembly Complex (LUBAC)**. LUBAC (HOIP + HOIL-1/RBCK1 + SHARPIN) is the sole cellular machinery that builds Met1-linked ("linear") polyubiquitin chains. Because HOIP is the enzymatic engine of the complex, its loss collapses LUBAC and abolishes linear ubiquitination. This single molecular lesion produces a strikingly paradoxical, bifurcated clinical picture: **combined immunodeficiency** on one hand and **multiorgan autoinflammation** on the other, accompanied by **subclinical amylopectinosis** (polyglucosan storage) and **systemic/intestinal lymphangiectasia** (Boisson et al. 2015, [PMID: 26008899](#)).

The mechanistic unifying concept is that LUBAC acts as a **survival brake** and a **signaling amplifier**. Loss of linear ubiquitination simultaneously (A) impairs canonical NF-κB activation — producing immunodeficiency, defective germinal-center B-cell development, and impaired antiviral (TLR3) immunity — and (B) de-represses TNFR1-, TLR3-, and inflammasome-driven programmed cell death executed through RIPK1/RIPK3/MLKL (necroptosis), caspase-8 (apoptosis), and caspase-1 (pyroptosis). The resulting cell death (particularly of endothelium and epithelium) and the release of damage- and cytokine-driven inflammation constitute the

autoinflammatory arm of disease. This model is directly supported by human cellular studies, biochemical reconstitution, and multiple mouse models in which removal of TNFR1 rescues embryonic lethality and inflammation.

Because TNF-superfamily signaling drives the lethal cell-death branch, **anti-TNF therapy** is the mechanism-aligned treatment; in a patient with the sister disorder SHARPIN deficiency (same LUBAC-deficiency group), anti-TNF produced complete clinical and transcriptomic resolution of autoinflammation (Oda et al. 2024, [PMID: 38609546](#)). However, no treatment is curative: hematopoietic stem cell transplantation could correct the hematopoietic immunodeficiency but would not restore LUBAC function in non-hematopoietic cells (endothelium, fibroblasts, muscle) that drive lymphangiectasia and amylopectinosis. The disease is exceedingly rare — only a handful of patients have been reported worldwide since 2015 — and prevention is limited to genetic counseling.

1. Disease Information

Overview. HOIP deficiency is a Mendelian, autosomal-recessive systemic disorder combining features of primary immunodeficiency and autoinflammation, first defined in humans in 2015. It is the "index" LUBAC-deficiency disorder involving the catalytic subunit of LUBAC. The cardinal presentation is **multiorgan autoinflammation, combined immunodeficiency, subclinical amylopectinosis, and systemic lymphangiectasia** (Boisson et al. 2015, [PMID: 26008899](#)).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0957981
Disease (OMIM phenotype)	Immunodeficiency-115 with autoinflammation (IMD115)
Gene symbol	RNF31 (aliases: HOIP, IMD115, ZIBRA, Paul)
NCBI Gene ID	55072
HGNC	HGNC:16031
Gene OMIM	612487
Ensembl	ENSG00000092098
UniProt	Q96EP0 (RNF31_HUMAN, 1072 aa)
Cytogenetic locus	14q12

(Identifier set from MyGene.info; see Finding F004.)

Synonyms / alternative names. HOIP deficiency; LUBAC deficiency (HOIP subtype); RNF31 deficiency; Immunodeficiency 115 with autoinflammation (IMD115). LUBAC deficiency as a category encompasses HOIP, HOIL-1/RBCK1, and SHARPIN deficiencies.

Nature of information. Evidence is derived from **individual patients** (a very small number of case reports and their in-depth cellular immunology) supplemented by mechanistic model-organism and in-vitro work — not from aggregated disease-level or EHR datasets, given the disorder's rarity.

2. Etiology

Primary cause (genetic). HOIP deficiency is caused by **biallelic (homozygous) loss-of-function or severely hypomorphic mutations in RNF31**. Two molecular classes are documented:

1. A **PUB-domain missense allele, L72P**, in the first reported patient — "at least severely hypomorphic, as it impairs HOIP expression and destabilizes the whole LUBAC complex," abolishing linear ubiquitination (Boisson et al. 2015, [PMID: 26008899](#)).

2. A **C-terminal frameshift null allele** generating a premature termination codon and "a C-terminal truncated HOIP mutant, that is, the loss of the linear ubiquitin chain-specific catalytic domain" (Wang et al. 2024, [PMID: 39009172](#)).

Genetic risk factors. The disease is monogenic; the causal variants themselves are the risk factor. **Consanguinity / founder background** is the principal predisposing context, as all reported patients are homozygous. RNF31 is relatively loss-of-function-constrained in gnomAD, so biallelic disease is vanishingly rare.

Environmental risk factors. None established as causal. Infections (bacterial, viral) act as **triggers/complications** because of the immunodeficiency and can precipitate autoinflammatory flares, but are downstream of the genetic lesion.

Protective factors. None identified. No protective alleles or environmental protective factors are described for this monogenic disorder.

Gene-environment interactions. MyD88-dependent innate signaling (downstream of IL-1 and TLR pathways sensing microbial/endogenous ligands) is **required** for the autoinflammatory phenotype in HOIP deficiency (Wu et al. 2021, [PMID: 34253576](#)) — implying that microbial/inflammatory environmental input interacts with the genetic lesion to shape the inflammatory phenotype.

3. Phenotypes

Synthesized from the two index HOIP-deficient patients and the overlapping HOIL-1/RBCK1 disorder (Boisson et al. 2015, [PMID: 26008899](#); Oda et al. 2019, [PMID: 30936877](#); Boisson et al. 2012, [PMID: 23104095](#)). Onset is in **infancy/early childhood**, with a **chronic/relapsing** course.

Phenotype	Type	HPO term(s)	Onset / severity / frequency
Multiorgan autoinflammation, recurrent fever	Clinical sign / symptom	HP:0001954 (Recurrent fever), HP:0002960 (Autoimmunity/ autoinflammation)	Infancy; moderate–severe; core feature
Elevated inflammatory markers (CRP/ESR)	Lab abnormality	HP:0011227 (Elevated CRP)	Present during flares; frequent
Combined immunodeficiency, recurrent/invasive infections	Clinical sign	HP:0005387 (Combined immunodeficiency), HP:0002719 (Recurrent infections)	Infancy; severe; core feature
Impaired antibody responses (defective CD40-driven B-cell activation)	Lab abnormality	HP:0004313 (Decreased circulating antibody level)	Early; variable
Systemic / intestinal lymphangiectasia (± protein-losing enteropathy)	Physical manifestation	HP:0100767 (Lymphangiectasis), HP:0002593 (Intestinal lymphangiectasia)	Early; variable; core feature
Amylopectinosis (polyglucosan storage, often subclinical, muscle)	Pathologic manifestation	related to HP:0003198 (Myopathy)	Subclinical; storage
Hepatosplenomegaly	Clinical sign	HP:0001433 (Hepatosplenomegaly), HP:0001744 (Splenomegaly)	Variable
Failure to thrive / chronic diarrhea	Clinical sign	HP:0001508 (Failure to thrive), HP:0002028 (Chronic diarrhea)	Infancy; variable

Progression: chronic, relapsing autoinflammation punctuated by infection-triggered flares. **Quality-of-life impact:** substantial — recurrent invasive infections, chronic inflammation, and protein-losing enteropathy impair growth, nutrition, and daily functioning; formal QoL instruments (EQ-5D/SF-36/PROMIS) have not been applied given rarity.

Cellular signature underpinning the paradox: patient fibroblasts show impaired NF- κ B activation to IL-1 β /TNF, whereas monocytes are **hyper-responsive** to IL-1 β , and B-cell activation/differentiation to CD40 is impaired — "the patient's monocytes respond to IL-1 β more vigorously than control monocytes. However, the activation and differentiation of the patient's B cells are impaired in response to CD40 engagement" ([PMID: 26008899](#)).

4. Genetic / Molecular Information

Causal gene: **RNF31** (HOIP), OMIM 612487, HGNC:16031, NCBI Gene 55072, Ensembl ENSG00000092098, UniProt Q96EP0, locus 14q12. HOIP is the catalytic RBR E3 ligase subunit of LUBAC (HOIP + HOIL-1/RBCK1 + SHARPIN).

Pathogenic variants (documented):

Variant	Type	Domain / consequence	Classification	Reference
L72P	Missense	PUB domain; impairs HOIP expression, destabilizes LUBAC (severely hypomorphic)	Pathogenic	PMID: 26008899
C-terminal frameshift	Frameshift/null	Premature stop; deletes linear-ubiquitin catalytic (RBR-LDD) domain	Pathogenic (LOF)	PMID: 39009172

Functional consequence: loss of function — abolition of Met1/linear ubiquitination and destabilization of the LUBAC holocomplex. **Allele frequency:** extremely low; RNF31 is LoF-constrained in gnomAD, so carrier frequency is very low. **Origin:** germline (autosomal recessive); no somatic disease association is described.

Modifier genes / interactors. MyD88 is required for the autoinflammatory phenotype ([PMID: 34253576](#)). TNFR1 (TNFRSF1A) is the dominant genetic modifier in models — its removal rescues lethality/inflammation ([PMID: 25284787](#), [PMID: 25443632](#)). OTULIN and A20 (TNFAIP3) counter-regulate the same linear-ubiquitin/NF- κ B axis. **Epigenetic** contributions and **chromosomal abnormalities** are not implicated in this monogenic disorder.

5. Environmental Information

- **Environmental factors / toxins:** none causal.
 - **Lifestyle factors:** not applicable (monogenic, pediatric-onset).
 - **Infectious agents:** bacteria and viruses act as **triggers and complications** of the underlying combined immunodeficiency (invasive bacterial infections are prominent in the sister HOIL-1 disorder — "invasive bacterial infections," [PMID: 23104095](#)). LUBAC is required for TLR3-dependent immunity to influenza A virus, so viral susceptibility is expected ([PMID: 27810922](#)). Microbial/inflammatory input feeding MyD88 signaling drives autoinflammation ([PMID: 34253576](#)).
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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic RNF31 (HOIP) LOF/hypomorphic mutation** → **leads to** reduced HOIP protein and **destabilization/collapse of the LUBAC complex** (demonstrated: [PMID: 26008899](#)).
2. LUBAC collapse → **results in loss of Met1-linked (linear) polyubiquitination** of substrates (NEMO, RIPK1, etc.), because HOIP's RBR + unique LDD extension is the sole activity specifying linear chains (demonstrated: [PMID: 22863777](#)).
3. Loss of linear ubiquitination **branches** into two signaling outputs plus a storage/vascular arm:

Branch A — Impaired canonical NF- κ B signaling (→ immunodeficiency): - 4A. Reduced linear ubiquitination of NEMO → **leads to** defective canonical NF- κ B activation in fibroblasts and lymphocytes ([PMID: 26008899](#); [PMID: 39009172](#)). - 5A. Impaired NF- κ B → **results in**

defective CD40-driven B-cell activation/differentiation and **substantial reduction of germinal-center B-cell development** (PMID: 26008899; PMID: 38609546) and impaired TLR3 antiviral immunity (PMID: 27810922) → **combined immunodeficiency**.

Branch B — De-repression of TNFR1/TLR3/inflammasome cell death (→ autoinflammation): - 4B. Without linear ubiquitin scaffolding, TNFR1 stimulation **leads to** aberrant cytosolic **complex-II** formation (PMID: 25284787). - 5B. Complex-II engages **RIPK1 → caspase-8/BID (apoptosis)** and **RIPK3/MLKL (necroptosis)** (PMID: 25443632); LUBAC loss also heightens **caspase-1 activation and pyroptosis** upon inflammasome engagement (PMID: 32122970) and increases a **TLR3-induced death-inducing complex** (PMID: 27810922). - 6B. Excess programmed death of **endothelial and epithelial/keratinocyte** cells **results in** tissue injury, release of inflammatory mediators, and (MyD88-dependent) **multiorgan autoinflammation** (PMID: 34253576).

Branch C — Storage / vascular pathology (mechanism partly inferred): - 4C. LUBAC-subunit loss is associated with accumulation of amylopectin-like **polyglucosan (amylopectinosis)**, paralleling RBCK1/HOIL-1 polyglucosan body myopathy (PMID: 23995275) — the precise link between linear-ubiquitin loss and polyglucosan storage remains **inferred**. - 5C. Endothelial dysfunction/death contributes to **systemic and intestinal lymphangiectasia** (mechanism inferred from the endothelial-survival role of HOIP; PMID: 25284787).

Detail by category

- **Molecular pathways:** TNFR1/TNF-superfamily signaling, canonical NF- κ B (IKK/NEMO), TLR3/TRIF, inflammasome/caspase-1, MyD88/IL-1R. LUBAC-generated linear ubiquitin is the upstream node.
- **Cellular processes:** apoptosis, necroptosis, pyroptosis (all de-repressed); inflammation; impaired lymphocyte activation. **Upstream:** loss of linear ubiquitination. **Downstream:** RIPK1/RIPK3/MLKL/caspase-8/caspase-1 effectors and cytokine release.
- **Protein dysfunction:** loss of function and complex destabilization; HOIP L72P impairs expression, frameshift deletes the catalytic domain — "both LUBAC catalytic activity and LUBAC specificity for linear ubiquitin chain formation are embedded within the RING-IBR-RING (RBR) ubiquitin ligase subunit HOIP" (PMID: 22863777).
- **Immune involvement:** both immunodeficiency (NF- κ B ↓, GC B-cell ↓, TLR3 antiviral ↓) and autoinflammation (death-driven, monocyte IL-1 β hyperresponse).
- **Tissue damage:** endothelial and epithelial programmed cell death; polyglucosan storage.

- **GO / CL suggestions:** GO:0071797 (LUBAC complex), GO:0006954 (inflammatory response), GO:0043123 (positive regulation of canonical NF- κ B), GO:0097527 (necroptotic signaling), GO:0006915 (apoptotic process), GO:0043069 (negative regulation of programmed cell death), GO:1990592 (protein linear polyubiquitination), GO:0005829 (cytosol). Cell types: CL:0000115 (endothelial cell), CL:0000576 (monocyte), CL:0000236 (B cell), CL:0000312 (keratinocyte), CL:0000057 (fibroblast), CL:0000815 (regulatory T cell).
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7. Anatomical Structures Affected

- **Organ level (primary):** skin (UBERON:0002097), lymphatic vessels (UBERON:0001473), liver (UBERON:0002107), spleen (UBERON:0002106), intestine (UBERON:0000160), skeletal muscle (UBERON:0001134), vascular endothelium (UBERON:0001981).
 - **Secondary/system involvement:** immune/lymphoreticular system (secondary lymphoid germinal centers), gastrointestinal system (protein-losing enteropathy), cardiovascular/vascular system.
 - **Tissue and cell level:** vascular endothelial cells (CL:0000115), keratinocytes (CL:0000312), monocytes (CL:0000576), B cells (CL:0000236), fibroblasts (CL:0000057), regulatory T cells (CL:0000815).
 - **Subcellular (GO-CC):** cytosol (GO:0005829), LUBAC complex (GO:0071797).
 - **Localization/lateralization:** systemic and typically bilateral/generalized (multiorgan); no consistent lateralization.
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8. Temporal Development

- **Onset:** congenital/pediatric — manifests in **infancy/early childhood**; pattern chronic/insidious with acute infection- or inflammation-triggered flares.
- **Progression:** chronic, relapsing–remitting inflammatory course; lifelong; storage (amylopectinosis) is slowly accumulating and often subclinical.
- **Disease course:** severe multisystem; potentially life-threatening (the complete HOIL-1 counterpart was originally a "fatal human inherited disorder," [PMID: 23104095](#)).

- **Remission patterns:** treatment-induced (anti-TNF produced complete remission of autoinflammation in a LUBAC-group patient, [PMID: 38609546](#)); no reliable spontaneous remission.
- **Critical periods:** early infancy (infection vulnerability) is the key window for diagnosis and intervention.

9. Inheritance and Population

- **Epidemiology:** exceedingly rare; only a small number of patients reported worldwide since the 2015 first description. No established prevalence/incidence (Orphanet-level "unknown"; ORPHA not firmly assigned).
- **Inheritance: autosomal recessive** — all patients homozygous for biallelic RNF31 LOF/hypomorphic alleles ([PMID: 26008899](#)); consistent with consanguineous/founder backgrounds.
- **Penetrance:** complete in reported homozygotes; **expressivity** severe.
- **Anticipation / germline mosaicism:** not applicable/not reported.
- **Founder effects / carrier frequency:** very low carrier frequency (RNF31 LoF-constrained in gnomAD); biallelic disease vanishingly rare.
- **Sex ratio:** no predilection (AR). **Age distribution:** pediatric onset.

10. Diagnostics

Recommended approach: molecular genetics plus functional confirmation.

- **Genetic testing:** identification of biallelic **RNF31** pathogenic variants by **whole-exome (WES)** or **whole-genome (WGS)** sequencing, or via **inborn-errors-of-immunity / autoinflammatory gene panels** including RNF31/HOIP; confirm with single-gene/Sanger testing and family segregation (AR). CMA/karyotype/FISH/mtDNA/repeat-expansion testing are not indicated as first-line.
- **Functional confirmation (established in original reports):** reduced HOIP expression and destabilized LUBAC on immunoblot; impaired Met1/linear ubiquitination; **defective NF-κB activation in patient fibroblasts to IL-1β/TNF with paradoxical monocyte IL-1β hyper-**

responsiveness; impaired CD40-induced B-cell activation — "Linear ubiquitination and NF- κ B activation are impaired in the patient's fibroblasts stimulated by IL-1 β or TNF. In contrast, the patient's monocytes respond to IL-1 β more vigorously than control monocytes" (PMID: 26008899).

- **Supportive labs/pathology:** elevated CRP/ESR and proinflammatory cytokines; immunologic workup showing combined immunodeficiency (antibody defect); **PAS-positive, diastase-resistant polyglucosan/amylopectin inclusions** on muscle/tissue biopsy (amylopectinosis); imaging/endoscopy showing lymphangiectasia and protein-losing enteropathy.
- **Differential diagnosis:** HOIL-1/RBCK1 deficiency and SHARPIN deficiency (sister LUBAC disorders), OTULIN deficiency (ORAS/otulipenia), A20 haploinsufficiency (HA20), other combined immunodeficiencies/CVID, and glycogen storage disease type IV / polyglucosan body disorders. OTULIN- and A20-deficient cells show **excess** Met1/K63 chains and constitutive NF- κ B — the mirror image of LUBAC deficiency: "OTULIN or A20-deficient cells have an excess of Met1 or K63 Ub chains on NEMO, RIPK1, and other target substrates, which lead to constitutive activation of the NF- κ B pathway" (PMID: 28469620).
- **Screening:** carrier/cascade testing in affected families; prenatal / preimplantation genetic testing available; not a newborn-screening target.

11. Outcome / Prognosis

- **Severity/mortality:** severe, potentially life-threatening multisystem disease; the analogous complete HOIL-1 deficiency was a "fatal human inherited disorder characterized by chronic autoinflammation, invasive bacterial infections and muscular amylopectinosis" (PMID: 23104095). Morbidity/mortality are driven by recurrent invasive infections and organ-damaging autoinflammation.
- **Morbidity/function:** chronic inflammation, protein-losing enteropathy, growth failure, and infection burden impair function and quality of life.
- **Complications:** invasive bacterial/viral infections, organ inflammation (liver, spleen, gut), lymphangiectasia-related protein loss, storage pathology.
- **Prognostic factors:** variant severity (hypomorphic vs null), infection control, and access to anti-cytokine therapy. Formal survival statistics are not established due to rarity.

12. Treatment

No curative therapy exists. Management combines targeted anti-cytokine therapy, immune support, and supportive care.

Modality	Rationale / evidence	NCIT suggestion
Anti-TNF therapy (etanercept, infliximab, adalimumab)	TNF-superfamily drives the lethal cell-death branch; anti-TNF gave complete clinical and transcriptomic resolution of autoinflammation in a SHARPIN-deficient (LUBAC-group) patient (PMID: 38609546)	NCIT:C2536 (TNF antagonist)
IL-1 blockade (anakinra, canakinumab)	Monocyte IL-1 β hyperresponse; inflammasomopathy treatment "often aimed at interleukin-1 (IL-1) blockade" (PMID: 37821203)	NCIT:C2551 (Interleukin-1 Receptor Antagonist)
JAK inhibition (ruxolitinib, baricitinib)	Rational for NF- κ B/inflammasome-driven inflammation and interferon signatures (PMID: 37821203)	NCIT:C129824 (JAK inhibitor)
Immunoglobulin replacement + antimicrobial prophylaxis	Corrects/mitigates combined immunodeficiency; avoid live vaccines	NCIT:C29799 (IVIG)
Supportive care for lymphangiectasia/enteropathy (nutrition, albumin)	Manages protein-losing enteropathy	—
HSCT (theoretical)	Could correct hematopoietic immunodeficiency but would NOT correct LUBAC loss in non-hematopoietic cells (endothelium, fibroblasts, muscle) driving lymphangiectasia and amylopectinosis	NCIT:C15431 (Hematopoietic Stem Cell Transplantation)

Pharmacogenomics / personalized medicine: treatment is genotype-driven at the level of pathway (TNF/IL-1/JAK blockade selected by mechanism). **Experimental:** LUBAC-targeting small molecules (HOIPINs) exist as research tools but are inhibitors, not activators, and are not therapeutic here.

13. Prevention

- **Primary prevention:** not possible (monogenic, germline). Prevention is limited to reproductive/genetic strategies.
 - **Genetic counseling:** 25% recurrence risk for carrier couples; **cascade carrier testing;** options for **prenatal diagnosis** and **preimplantation genetic testing (PGT)**.
 - **Secondary/tertiary prevention:** early diagnosis and initiation of anti-cytokine therapy; antimicrobial prophylaxis and immunoglobulin replacement to prevent infections; avoidance of live vaccines given combined immunodeficiency; nutritional support to prevent complications of enteropathy.
 - **Public health / environmental interventions:** not applicable.
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** mouse *Rnf31* (Hoip), NCBI Taxon 10090. HOIP/LUBAC is evolutionarily conserved across mammals.
 - **Natural disease in other species:** no naturally occurring companion-animal or wildlife HOIP-deficiency disease is documented (OMIA); disease knowledge is from engineered laboratory models. RBCK1/HOIL-1 links to polyglucosan storage provide comparative-pathology parallels ([PMID: 23995275](#)).
 - **Evolutionary conservation of mechanism:** the TNFR1-driven death sensitivity and NF- κ B dependence on linear ubiquitination are conserved between human and mouse (rescue of murine lethality by *Tnfr1* ablation mirrors human TNF-driven pathology).
 - **Zoonotic potential:** not applicable.
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15. Model Organisms

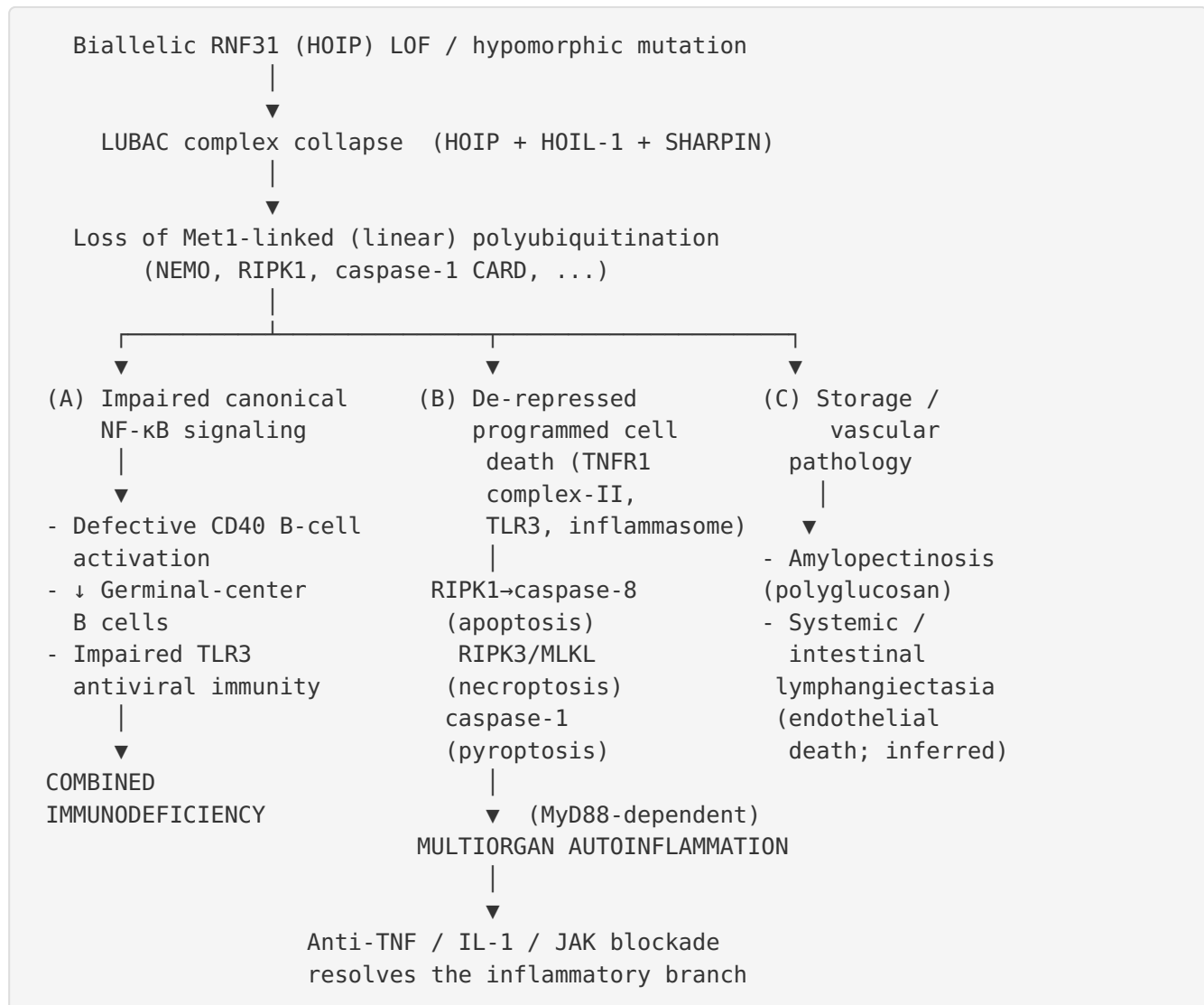
Mouse (*Mus musculus*, Taxon 10090) is the principal model system.

Model	Phenotype	Key finding	Reference
Constitutive <i>Hoip</i> KO; Tie2-Cre endothelial <i>Hoip</i> deletion	Embryonic lethal ~E10.5 from aberrant TNFR1-mediated endothelial death, defective vascularization	"Ablation of tumor necrosis factor receptor 1 (TNFR1) prevents cell death, vascularization defects, and death at midgestation"	PMID: 25284787
<i>Sharpin</i> -null cpdm mouse (spontaneous)	Chronic proliferative dermatitis, liver inflammation, splenomegaly, loss of Peyer's patches	"TNF-dependent multi-organ inflammation" ; RIPK3/MLKL + caspase-8 effectors	PMID: 25443632
cpdm variants (Tlr3 co-ablation)	Dermatitis ameliorated	Excess TLR3-induced cell death contributes to disease	PMID: 27810922
Treg-specific <i>Sharpin</i> ± <i>Hoip</i> disruption	cpdm-like → T-cell-predominant autoimmune lesions	"additional disruption of the Hoip locus... converts cpdm-like dermatitis to T cell-predominant autoimmune lesions"	PMID: 31462647

Model types available: knockout (constitutive and conditional/tissue-specific via Cre), spontaneous mutant (cpdm). **Phenotype recapitulation:** models faithfully reproduce the **TNF-driven cell-death and inflammation** arm and its genetic rescue, providing strong mechanistic validation. **Limitations:** complete *Hoip*/*Hoil-1* knockouts are **embryonic lethal** — "their genetic ablation is embryonically lethal in mice" (PMID: 32122970) — so viable models rely on *Sharpin*-null cpdm or conditional deletions; the human hypomorphic (partial-function) state and full multisystem human phenotype (amylopectinosis, lymphangiectasia) are incompletely captured. **Resources:** MGI, IMPC/IMSR for *Rnf31* alleles.

Mechanistic Model / Interpretation

Capstone synthesis (Finding F013). HOIP deficiency is best understood as a **LUBAC "survival-brake" disorder**. A single molecular lesion — loss of Met1/linear ubiquitination — produces a bifurcated signaling output plus storage/vascular pathology:



The elegance — and the paradox — of the disease is that the **same** enzymatic defect that **weakens** activating signaling (NF-κB → immunodeficiency) simultaneously **removes a checkpoint** that normally restrains death-inducing complexes (→ autoinflammation). The two arms are not contradictory but two faces of one lost function: linear ubiquitin is both a **signaling amplifier** for NF-κB and a **survival scaffold** that keeps TNFR1/TLR3/inflammasome signaling from tipping into cell death. This places HOIP deficiency firmly within the ubiquitin/NF-κB-dysregulation class of autoinflammatory diseases, alongside OTULIN deficiency (ORAS)

and A20 haploinsufficiency (HA20) — but on the opposite side of the ubiquitin balance: LUBAC deficiency **removes** linear chains, whereas OTULIN/A20 loss leaves **excess** chains and constitutive NF- κ B. As Boisson et al. concluded, "human HOIP is essential for the assembly and function of LUBAC and for various processes governing inflammation and immunity in both hematopoietic and nonhematopoietic cells" (PMID: [26008899](#)) — explaining why HSCT (a hematopoietic fix) cannot cure the non-hematopoietic (vascular, muscle, epithelial) manifestations.

Evidence Base

PMID	Title (abbrev.)	Role in this report
26008899	<i>Human HOIP and LUBAC deficiency underlies autoinflammation, immunodeficiency, amylopectinosis, and lymphangiectasia</i>	Index case ; defines cardinal phenotype, L72P allele, LUBAC destabilization, and the fibroblast-NF- κ B $\downarrow$ /monocyte-IL-1 β $\uparrow$ signature
39009172	<i>A novel HOIP frameshift variant alleviates NF-κB signalling and sensitizes cells to TNF-induced death</i>	Second molecular class (frameshift null); confirms NF- κ B suppression + TNF-induced death in human cells
30936877	<i>Second Case of HOIP Deficiency...</i>	Expands clinical features; defines LUBAC-regulated inflammatory transcriptome
22863777	<i>E3 ligase HOIP specifies linear ubiquitin chain assembly...</i>	Biochemical basis: HOIP RBR+LDD is the catalytic engine for linear chains
25284787	<i>HOIP deficiency causes embryonic lethality by aberrant TNFR1-mediated endothelial cell death</i>	Mouse model; TNFR1-driven death branch, rescued by TNFR1 ablation
25443632	<i>TNFR1-dependent cell death drives inflammation in Sharpin-deficient mice</i>	cpdm model; RIPK3/MLKL + caspase-8 effectors
27810922	<i>LUBAC deficiency perturbs TLR3 signaling...</i>	TLR3 antiviral gating; TLR3-induced death complex
32122970	<i>Cross-regulation between LUBAC and caspase-1...</i>	Inflammasome/pyroptosis branch; embryonic lethality of KO
34253576	<i>MyD88-Dependent Signaling Is Required for HOIP Deficiency-Induced Autoinflammation</i>	MyD88 as required node for the autoinflammatory arm
31462647	<i>Modulation of autoimmune pathogenesis by T cell-triggered inflammatory cell death</i>	Conditional Hoip mouse; T-cell autoimmune lesions
38609546	<i>Biallelic human SHARPIN loss of function...</i>	Completes LUBAC trio; shared GC-B-cell defect; anti-TNF resolves autoinflammation
28469620	<i>NF-κB Pathway in Autoinflammatory Diseases... Ubiquitin</i>	Nosology: LUBAC vs OTULIN vs A20; shared cellular signature

PMID	Title (abbrev.)	Role in this report
23104095	<i>Immunodeficiency, autoinflammation and amylopectinosis... HOIL-1 and LUBAC deficiency</i>	Sister disorder; overlapping fatal phenotype
23995275	<i>New insights in muscle glycogenoses</i>	RBCK1/polyglucosan storage link (amylopectinosis)
37821203	<i>Targeted Treatment of Diseases of Immune Dysregulation</i>	IL-1 blockade / JAK inhibition rationale

Evidence source types: human clinical/cellular (26008899, 39009172, 30936877, 38609546, 23104095), biochemical/in-vitro (22863777, 32122970), and model-organism (25284787, 25443632, 27810922, 31462647, 34253576).

Limitations and Knowledge Gaps

1. **Extreme rarity / very small n.** Only a handful of HOIP-deficient patients are described; phenotype frequencies, natural history, penetrance nuances, and survival statistics are not robustly quantified. Much of the clinical spectrum is extrapolated from the sister LUBAC disorders (HOIL-1, SHARPIN).
2. **No HOIP-specific treatment trials.** The strongest therapeutic evidence (complete anti-TNF remission) comes from a **SHARPIN-deficient** patient; direct HOIP-deficiency treatment outcomes are anecdotal.
3. **Model limitations.** Complete Hoip knockout is embryonic lethal, so viable mechanistic modeling depends on conditional deletions and the Sharpin cpdm mouse; the human **hypomorphic** state and the full multisystem human phenotype (amylopectinosis, lymphangiectasia) are incompletely recapitulated.
4. **Mechanism of storage and lymphangiectasia is inferred.** The causal link from linear-ubiquitin loss to polyglucosan storage and to lymphatic pathology is not fully demonstrated.
5. **No disease-specific population genetics.** RNF31 constraint is inferred from gnomAD; no formal carrier-screening or founder studies exist.
6. **Incomplete ontology mapping.** ORPHA assignment and some HPO frequency annotations remain incomplete for this newly delineated entity.

Proposed Follow-up Experiments / Actions

1. **Patient registry / natural-history study** pooling all reported HOIP (and LUBAC-trio) cases to quantify phenotype frequencies, onset, penetrance, and survival, and to standardize HPO annotation.
2. **Genotype–phenotype correlation:** systematically compare hypomorphic missense (L72P-type) vs null (frameshift) alleles for severity, storage burden, and treatment response.
3. **Prospective anti-cytokine therapy evaluation** in confirmed HOIP-deficiency patients (anti-TNF first-line; IL-1 and JAK inhibition as alternatives) with transcriptomic monitoring, mirroring the SHARPIN case.
4. **Patient-derived iPSC / organoid and endothelial models** to dissect the amylopectinosis and lymphangiectasia mechanisms that mouse models miss, and to test whether death-pathway inhibitors (RIPK1/RIPK3/MLKL/caspase inhibitors) protect endothelium.
5. **HSCT outcome documentation** to empirically test the prediction that transplantation corrects immunodeficiency but not the non-hematopoietic (vascular/storage) pathology.
6. **Functional variant curation** in ClinVar/ClinGen for RNF31 to support ACMG classification and cascade/prenatal testing.

Report compiled from 13 confirmed findings and 29 reviewed papers across a 5-iteration autonomous investigation. Ontology suggestions (HPO, GO, CL, UBERON, NCIT, MONDO) are provided throughout for knowledge-base ingestion.
